THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 6

1. At a party N men throw their hats into the center of a room. The hats are mixed up and each man randomly selects one. Let M be the number of men who select their own hat. Find $\mathbb{E}[M]$ and $\mathrm{Var}(M)$.

2. Suppose $X_1, X_2, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables with mean μ and variance σ^2. Let the sample mean be defined by

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

   (a) Find $\mathbb{E}[\bar{X}]$.

   (b) Find $\mathrm{Var}(\bar{X})$.

   (c) Find $\mathrm{Cov}(\bar{X}, X_i - \bar{X})$ for $i = 1, 2, \ldots, n$.

3. (a) Let $X_1, X_2, \ldots, X_n$ be i.i.d. Poisson random variables with parameters $\lambda_1, \lambda_2, \ldots, \lambda_n$, respectively. Prove that the sum $Y = X_1 + X_2 + \cdots + X_n$ is a Poisson random variable with parameter $\lambda = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.

   (b) Let $X_1, X_2, \ldots, X_n$ be i.i.d. exponential random variables with parameter λ. Prove that the sum $Y = X_1 + X_2 + \cdots + X_n$ is a gamma random variable with parameters n and λ.